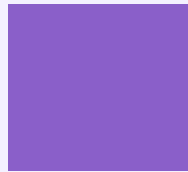


PRIMARY 5 SCIENCE

Coursebook — Term 3 Edition



UltraEdu Learning Press

This coursebook is original content created for the UltraEdu platform.

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Chapter 1: Living and Non-Living Things

All things around us can be sorted into two groups: living things and non-living things. Living things share a set of characteristics that non-living things do not have.

Characteristics of living things:

- They can grow and change in size.
- They need food, water, and air to survive.
- They can respond to things around them.
- They can reproduce to make more of their own kind.
- They can move on their own, either the whole body or parts of it.

Try it yourself:

Look around your classroom. List three living things and three non-living things you can see. For each living thing, name one characteristic that shows it is alive.

LIVING

e.g. a plant, a bird, a fish

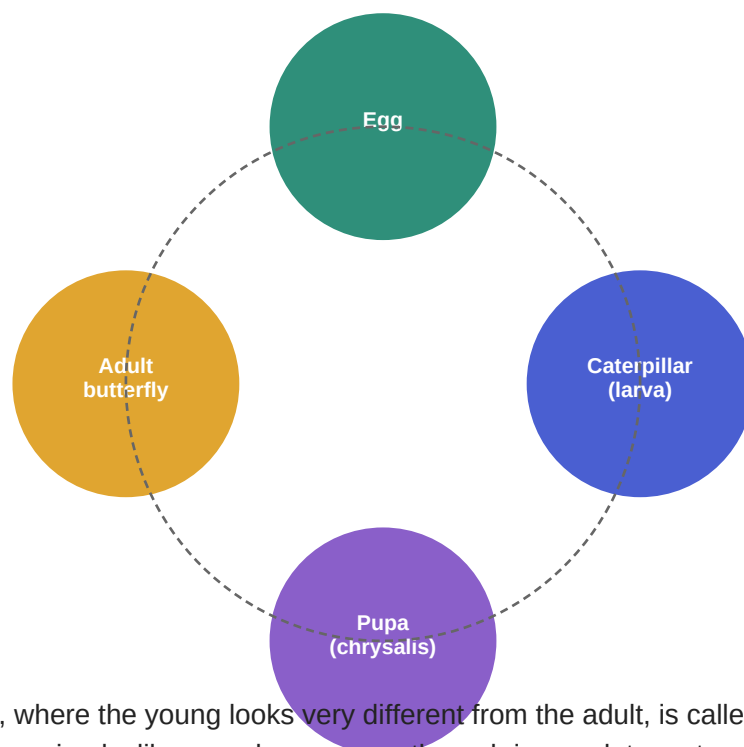
NON-LIVING

e.g. a rock, a chair, water

Chapter 2: Life Cycles of Animals

A life cycle shows the stages an animal goes through as it grows from the start of its life to when it can have young of its own. Different animals have different life cycles.

Example: Life cycle of a butterfly

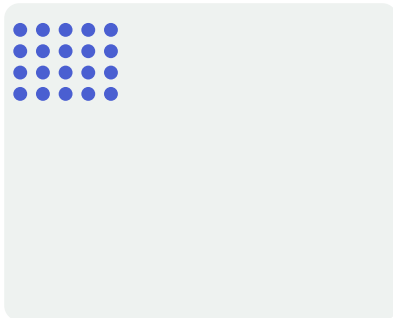


This kind of life cycle, where the young looks very different from the adult, is called complete metamorphosis. Other animals, like grasshoppers, go through incomplete metamorphosis, where the young already looks like a small version of the adult.

Chapter 3: States of Matter

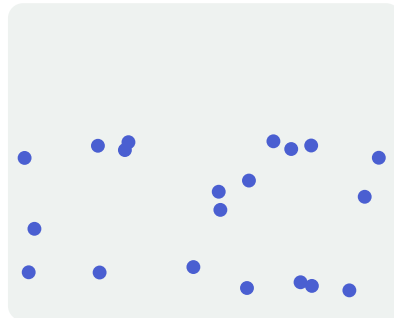
Matter is anything that takes up space and has mass. Matter can exist in three states: solid, liquid, and gas. The particles in each state are arranged differently.

SOLID



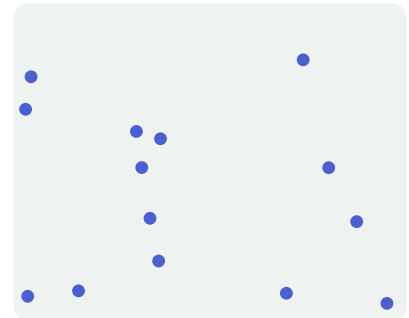
Particles packed tightly, fixed shape

LIQUID



Particles close but can move, takes shape of container

GAS



Particles spread far apart, fills any container

Changing state

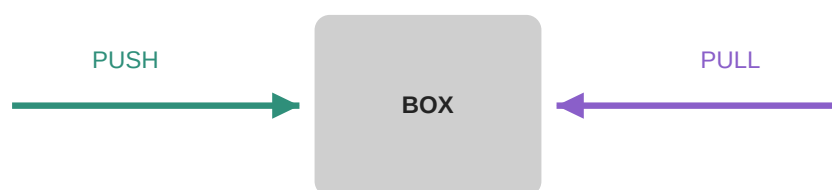
Matter can change from one state to another when it is heated or cooled. For example, ice (solid) melts into water (liquid) when heated, and water freezes into ice when cooled.

Chapter 4: Forces and Energy

A force is a push or a pull. Forces can make an object start moving, stop moving, speed up, slow down, or change direction. Forces can also change the shape of an object.

Common types of forces:

- Gravitational force — pulls objects towards the Earth.
- Frictional force — acts between two surfaces in contact, opposing motion.
- Magnetic force — a push or pull between magnets and certain metals.
- Elastic force — found in stretched or compressed springs and rubber bands.



Energy is needed to make a force act on an object. When you push a box, you use energy from your muscles. Machines use energy from fuel or electricity to produce forces that do work.

Chapter 5: The Water Cycle

The water cycle describes how water moves between the Earth's surface and the atmosphere. It has no beginning or end — it is a continuous cycle made up of several stages.

1

Evaporation

The sun heats water in the sea, rivers and lakes, turning it into water vapour.

2

Condensation

Water vapour rises, cools, and forms tiny water droplets which make clouds.

3

Precipitation

When clouds get heavy with water droplets, water falls as rain, hail or snow.

4

Collection

Rain collects in seas, rivers, lakes and underground, ready to evaporate again.

Chapter Review Questions

1. Name three characteristics that all living things share.
2. Draw and label the four stages of a butterfly's life cycle.
3. Give one example each of a solid, a liquid, and a gas found in your kitchen.
4. Describe two ways a force can change how an object moves.
5. Put the four stages of the water cycle in the correct order.

Glossary

Matter:	Anything that takes up space and has mass.
Force:	A push or a pull acting on an object.
Evaporation:	The process of a liquid changing into a gas.
Metamorphosis:	The process of changing form during a life cycle.